ARCHIE BENN

HOW TO TRAIN YOUR MODEL

ASSESSING THE IMPACTS OF FOREST AGE ON THE SPATIAL AND TEMPORAL GENERALISATION OF RANDOM FOREST EVAPOTRANSPIRATION MODELS

MSc Bioinformatics
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ABSTRACT

The ability to accurately predict actual evapotranspiration (ET) in regions using methods requiring minimal data input would enable estimations to be made in areas which are underrepresented globally due to the high costs involved in direct ET measurements. These estimations could, in turn, aid in more efficient use of freshwater and in predicting natural phenomena such as droughts. The development of ML models to predict ET has gained popularity in recent years, and being able to understand what links exist between a training dataset and a model which performs well on out-of-sample data should aid in the formation of better-predicting models, thus any explanatory variables which may aid in ET estimations should be explored. One under explored feature in ET ML models is forest age, which may impact ET rates due to age-related changes to canopy complexity and hydraulic conductance. The present study considers the use of forest age as a feature in ML predictions of ET at forest sites contained within the FLUXNET by using methods to explore spatial generalisation of models trained with and without age included, alongside temporal generalisation to assess whether forest age may help predictions across time at single sites. Results from spatial generalisation suggest nuanced results, with age benefiting predictions at some held out test sites, and hurting at others. Further investigation into two sites with opposing results pointed towards x being a factor (MAYBE?). Temporal generalisation suggested forest age x y z. These findings suggest that forest age may be useful to consider in ML ET modelling in the future, however considerations relating to tree species and the impacts of age on tree dynamics within each species may be useful in determining whether to use forest age or not in predictions. Suggestions for future work include accurate age surveys to be carried out at other FLUXNET sites which may permit 'per-species models' to be formed and tested for use with forest age.

Key words: Machine Learning, Evapotranspiration, FLUXNET, Forest Age.
Word count: 1397

DEDICATION AND ACKNOWLEDGMENTS

DECLARATION

I declare that the work in this dissertation was carried out in accordance with the requirements of the University's Regulations and Code of Practice for Taught Programmes and that it has not been submitted for any other academic award. Except where indicated by specific reference in the text, this work is my own work. Work done in collaboration with, or with the assistance of others, is indicated as such. I have identified all material in this dissertation which is not my own work through appropriate referencing and acknowledgement. Where I have quoted or otherwise incorporated material which is the work of others, I have included the source in the references. Any views expressed in the dissertation, other than referenced material, are those of the author.

Archie Benn
August 21, 2026

CONTENTS

1 AIMS AND OBJECTIVES

The overarching aim of this research project is to investigate how the age of a forest may impact predictions of actual evapotranspiration rates (hereafter ET) using machine learning (ML) models in forest ecosystems. The stand age of a forest has been shown to impact ET dynamics [1], and therefore could alter how ML models form predictions of ET if age is included as a feature during training and inference.

Given the importance of predicting ET [2, 3] and the lack of infrastructure to measure ET directly with eddy covariance (EC) methods in many ecoregions [4], the first part of this project will focus on spatial generalisation, asking: *"If a model is built using data from one set of sites, how well can it form predictions of ET rates at a geographically unseen site, and does using forest age help?"*. To address this question, FLUXNET sites with forests aged by Besnard et al. [5] are used alongside a framework developed to quantify site similarities based on characteristics in section 2.1. Random Forest (RF) models are then trained with and without forest age as a feature, and predictions of ET rates from both directly compared at unseen sites, with ground-truth validation provided by EC measurements in section 2.2.

Estimating future ET rates could also be useful in shaping policies and water use management over time at a single location. The second part of this project aims to address the temporal generalisation of ET models with and without age as a feature, asking *"If a model is built using historic data, does knowledge of any existing age-ET relationship aid in the accuracy of future ET predictions?"*. To address this question, site-specific models are trained with and without age as a feature on the earliest 80% of site data, and tested against the most recent 20% using ground-truth validation provided by EC measurements. Time series decomposition can then be used to identify any underlying shapes in the form of these ET predictions over time, with both models compared in section 2.3.

The objectives of this project are threefold: To assess the use of forest age in models for (1) spatial and (2) temporal predictions of ET, and to (3) outline future directions for research in utilising forest age in ET predictions.

2 RESULTS

2.1 Flux Tower Site Clustering

Cluster analysis was carried out in order to classify sites into groups based on their characteristics. This classification permits the predictability of ET, both within these characterised groups and across them, to be investigated. Gower's distances, a measure between pairs of data points, from 0 = completely dissimilar to 1 = completely similar [6] (see section 6.2 for details), were computed using the features listed in Table 2 and saved as a matrix. This matrix was passed to the Partitioning Around Medoids (PAM) algorithm [7] with the silhouette method to form clusters and identify medoid sites - those which have minimal dissimilarity to others within each cluster and thus can be seen as the representative sites of each cluster.

The silhouette method identified $k = 13$ as the optimal number of clusters based on average silhouette width, but with only 50 sites present this value was dropped to $k = 6$ in order to yield sufficiently well-populated clusters across a range of site characteristics. The cluster assignment solution from PAM with $k = 6$ clusters resulted in a range of site counts by cluster from 3 sites (cluster 6) up to 18 sites (cluster 3), and can be seen in a cluster plot of the first two principal components (Figure 5a). The overall geographic distribution of clusters appears to follow an expected pattern with regards to climatic groups; for example, sites at high latitudes tended to be grouped in cluster 3, with Mediterranean and coastal sites typically grouped in cluster 5 (Figure 5b). This expected pattern of clusters suggests that the PAM clustering solution captures climate-relevant patterns to some degree, although this was not formally tested. The six medoid sites were geographically distributed with four sites in North America and two sites in Europe (Figure 5b), and a range of site ages from 20 years (US-Me3) to 117 years (DE-Lnf).

To define cluster site characteristics, the means of z-score standardised numerical features (mean = 0, sd = 1) and mode of categorical features were considered for each cluster. Mean scaled numeric features can be seen within a heat map (Figure 6) for a visual comparison across clusters. Mean cluster features ≥ 1 sd from the all site mean were considered as 'high/low' (≥ 1.5 sd as 'very high/very low') in order to qualitatively classify cluster characteristics, with these summarised in Table 1.

2.2 Spatial Generalisation

2.2.1 *Analysis 1.1 - Model Sensitivity*

Results from model sensitivity testing indicate mixed model performance across the medoid sites, based on regression metrics comparing the predicted and observed daily ET Figure 7. In this part of the analysis, features were sequentially added to RF models, and these were then re-trained with each additional feature before testing on all medoid sites (see section 6.3.1 for details).

Predictions of daily ET formed for CA-Obs, US-UMd, and DE-Lnf showed strong performance across the range of input features, while CA-TP3, IT-SRo, and US-Me3 performed poorly when considering R^2, RMSE and MAE.

CA-Obs and DE-Lnf showed very similar patterns of performance, with predictions at both sites climbing across metrics up until `Lai_500m` was added (Peak CA-Obs: $R^2 = 0.90$, RMSE = 0.28, MAE = 0.19; Peak DE-Lnf: $R^2 = 0.87$, RMSE = 0.42, MAE = 0.31). Predictions at both sites then declined with the inclusion of `Site_age` to the model. US-UMd deviated from this performance pattern slightly, with metrics improving further when `Site_age` was included in the model (Peak US-UMd: $R^2 = 0.87$, RMSE = 0.46, MAE = 0.34). At all three of these sites the inclusion of only a single feature in the RF model, `T_air`, demonstrated strong to moderate predictive performance (CA-Obs: $R^2 = 0.76$, RMSE = 0.42, MAE = 0.31; US-UMd: R^2 = 0.64, RMSE = 0.78, MAE = 0.58; DE-Lnf: R^2 = 0.63, RMSE = 0.73, MAE = 0.58).

Predictive performance for daily ET at IT-SRo showed similar patterns to those made at CA-Obs and DE-Lnf, with metrics generally improving up until a peak with the addition of `Lai_500m`, but with significantly reduced values (Peak IT-SRo: R^2 = 0.28, RMSE = 0.91, MAE = 0.67). Performance at the remaining two sites, CA-TP3 and US-Me3, was particularly mixed and relatively poor (Peak CA-TP3: R^2 = 0.54, RMSE = 0.71, MAE = 0.51; Peak US-Me3: R^2 = 0.0, RMSE = 0.75, MAE = 0.56). R^2 fluctuated between predictions made with consecutively added features at these two sites, with performance at US-Me3 indicating that the model has no explanatory power at this site based on an R^2 maximum of 0.0.

Ratios of RMSE:MAE from peak predictive models at each medoid all fall between 1.33 and 1.48, suggesting that the overall impact of outlier ET predictions, and therefore overall error distribution, was fairly consistent from the best performing models at all sites during this analysis.

2.2.2 *Analysis 1.2 - Sampled Training Sets*

Figure 1 shows the paired results between the 'Mod' and the 'Mod + Age' models (described by Equation 1 and Equation 2 respectively) during the shuffled and sampled training set analysis (see section 6.3.2 for details).

Results from this analysis indicate that including age had a significant impact across a range of training set site compositions, with all six medoid sites showing significant differences in R^2 between the 'Mod' and the 'Mod + Age' predictions for daily ET (Wilcoxon signed-rank tests, $p < 0.01$ for all). The greatest absolute ΔR^2 between the two models was at CA-TP3, with 91/100 runs resulting in an increased R^2 when age was added (mean ΔR^2 = +0.08), with a consistent spread of variability across the runs between the models (R^2_{Mod} = 0.46 ± 0.07; $R^2_{Mod+Age}$ = 0.54 ± 0.07). De-Lnf was the most negatively impacted of all medoid sites when age was added, with 91/100 runs resulting in a reduced R^2 (mean ΔR^2 = -0.05), and a five fold increase in variability across the runs (R^2_{Mod} = 0.86 ± 0.01; $R^2_{Mod+Age}$ = 0.81 ± 0.05).

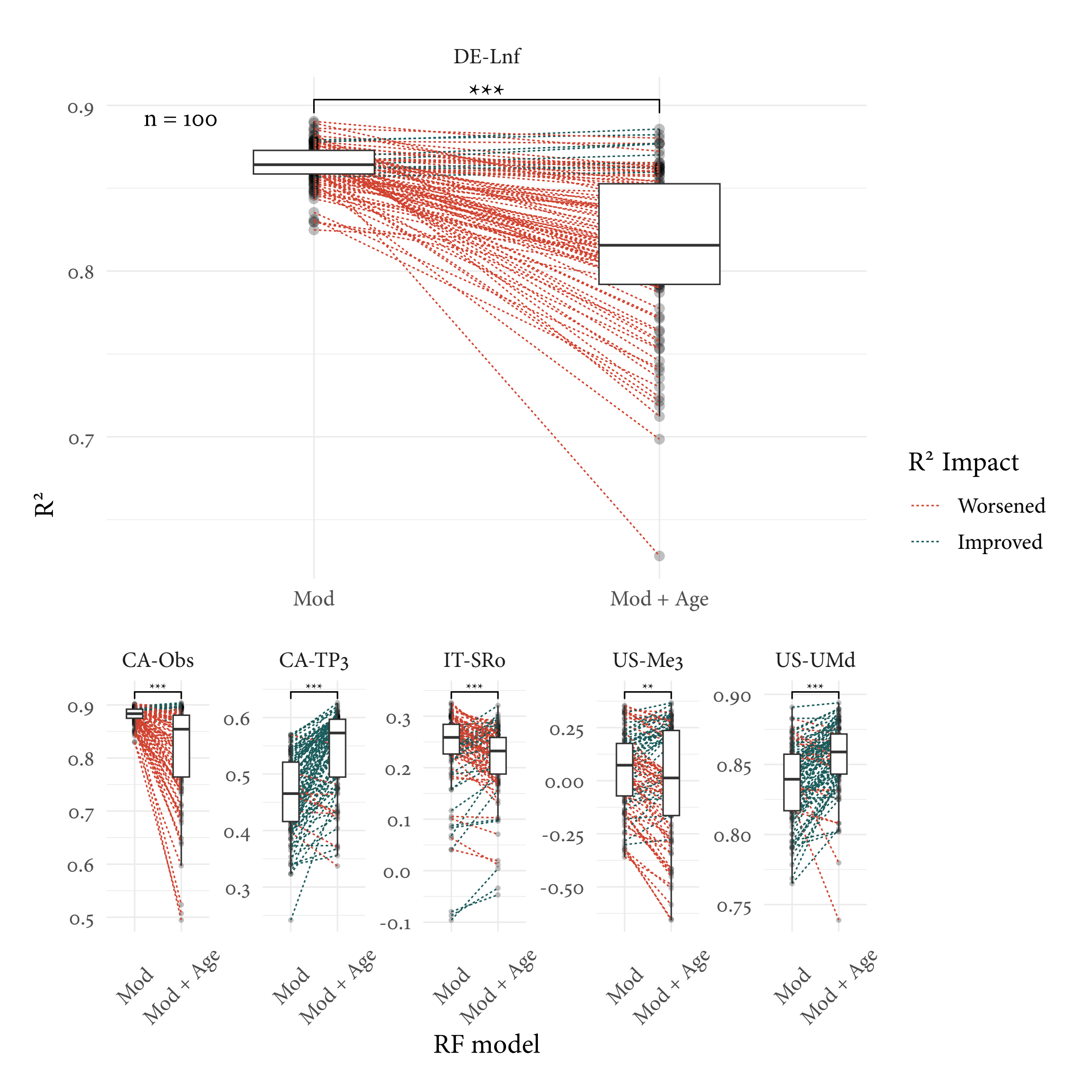

Figure 1: Paired box-plots indicate the impact on R^2 when age is added as a feature to RF models across 100 repeated runs of randomly sampled training datasets. Points are connected on a per-run basis where training dataset site composition is identical. Wilcoxon signed-rank tests suggest model performances differ significantly at all test sites, as indicated by the significance annotation on each facet (*** signifies $p < 0.001$, ** signifies $p < 0.01$ [8]). DE-Lnf is shown large for readability, with other cluster medoid sites shown at a reduced scale for comparison.

TRAINING SET COMPOSITIONS Here I will quickly look into the training set compositions of the RF models where age improved OR worsened R^2.

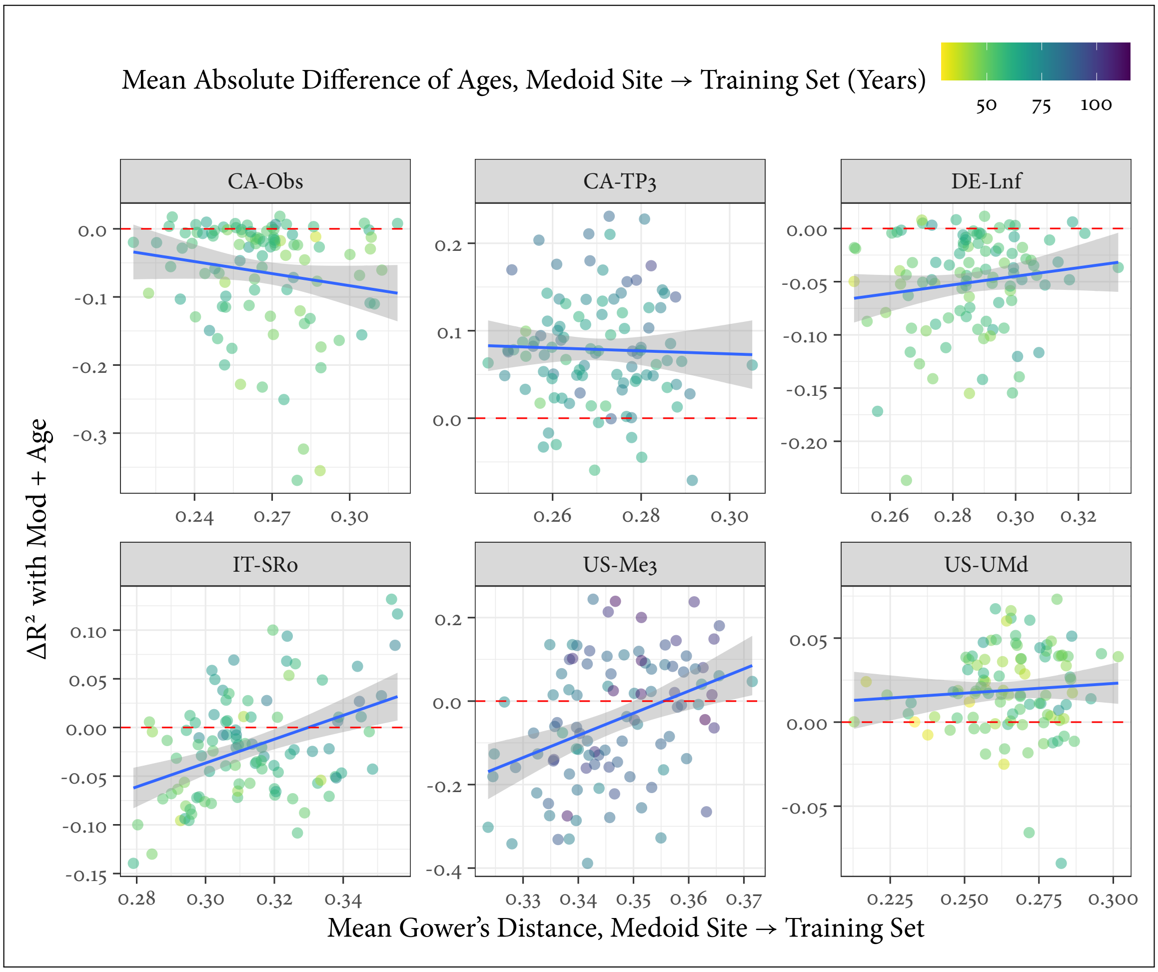

Figure 2: Plots

2.2.3 *Analysis 1.3 - Training by Cluster*

2.2.4 *Case Study of US-UMd and DE-Lnf*

- Look into predictions vs observed at both this R2 plot but also over timeframe
- US-UMd: Continental Site, 2715 days of data, 90 years old
- DE-Lnf: Temperate Site, 2670 days of data, 117 years old

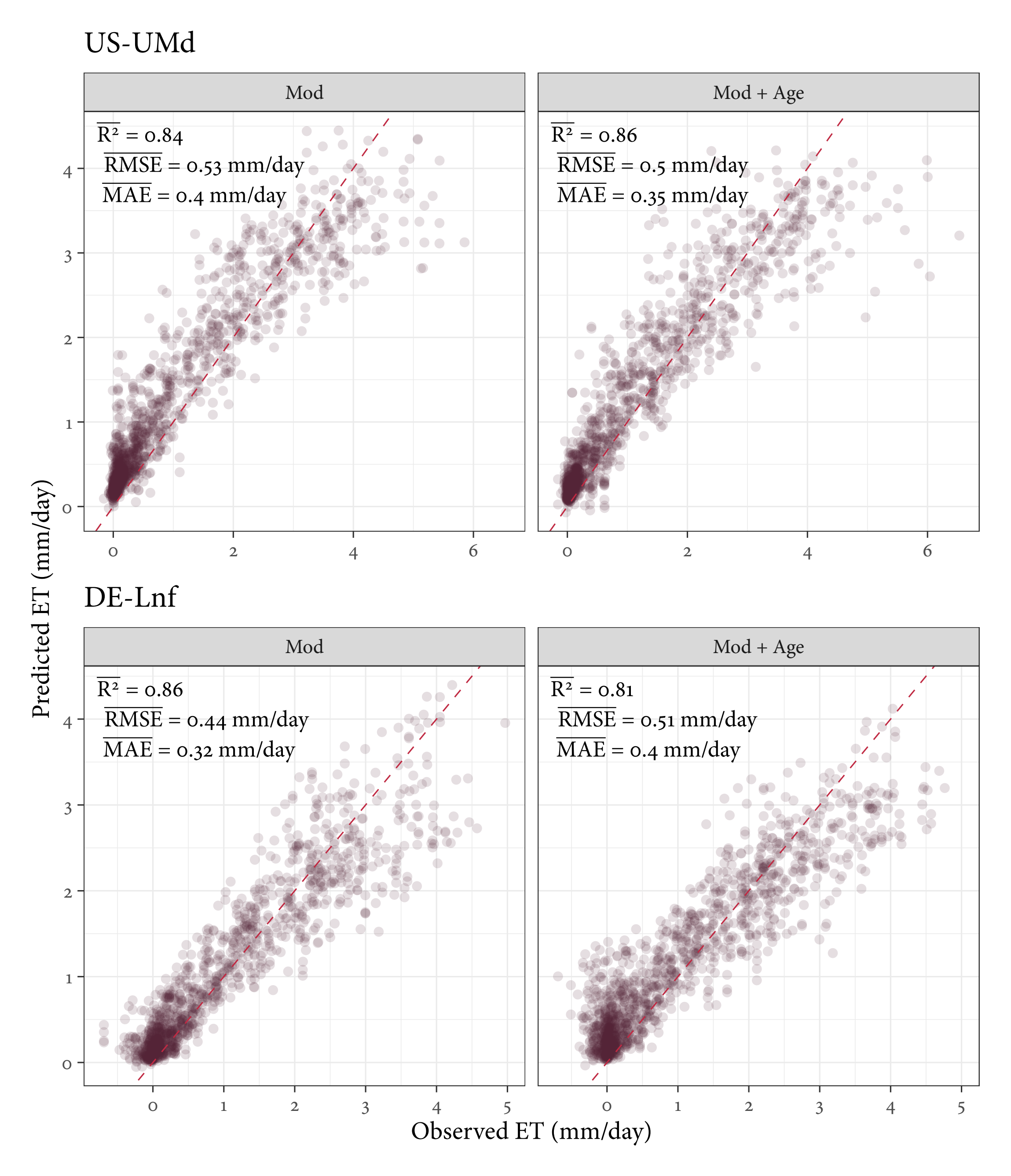

Figure 3: Subset of 2500 data points

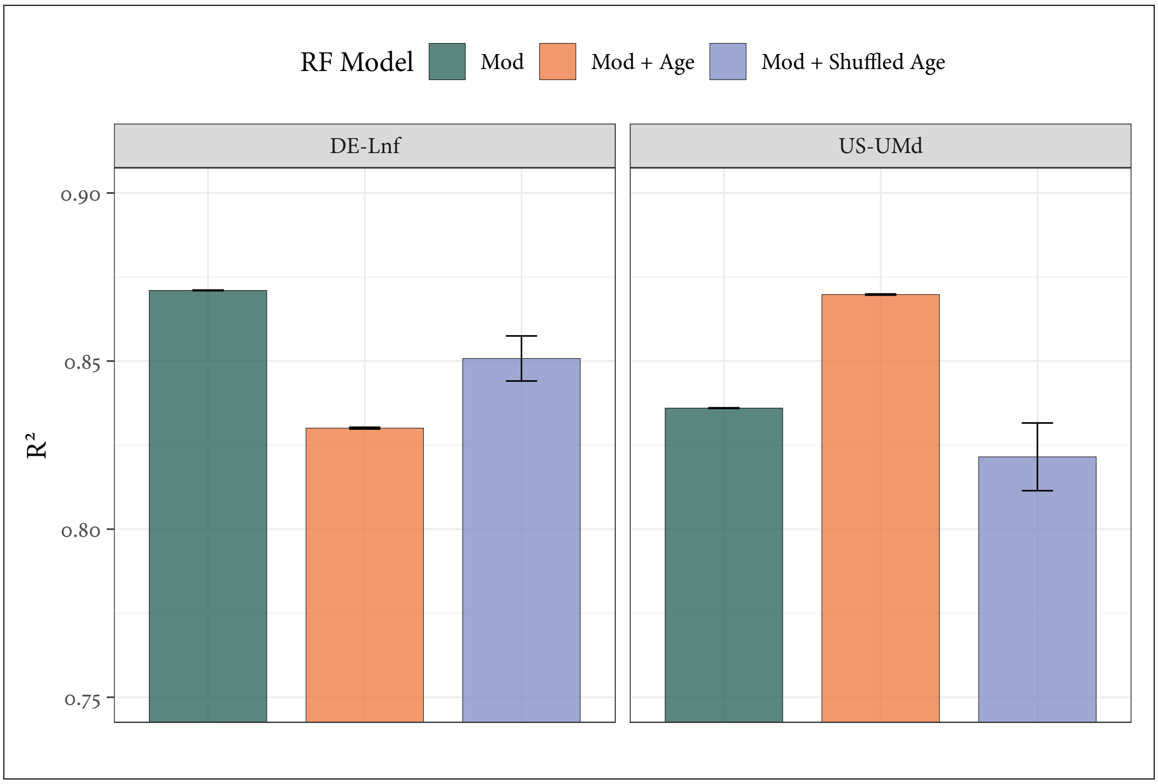

Figure 4: Bar plot showing the effect of different model setups on R^2 at DE-Lnf and US-UMd, with error bars indicating standard error. Models were trained using a leave one group out (LOGO) approach, with ten repeats using different `NumPy` seeds. To assess whether site age was acting as a proxy for other site-level characteristics in these models, a model which randomly shuffled ages between sites was also formed using identical seeds and LOGO approach. This 'shuffled age' model appeared to increase R^2 at DE-Lnf compared to the Mod + Age model where an opposite effect was observed at US-UMd, suggesting that true age-ET relationships learned by the model transfer well to predicting ET at US-UMd, but not at DE-Lnf.

2.3 Temporal Generalisation

2.3.1 *Analysis 2.1 - Prediction Accuracy*

2.3.2 *Analysis 2.2 - Assessing Trends with Time Series Decomposition*

3 DISCUSSION

4 CONCLUSION

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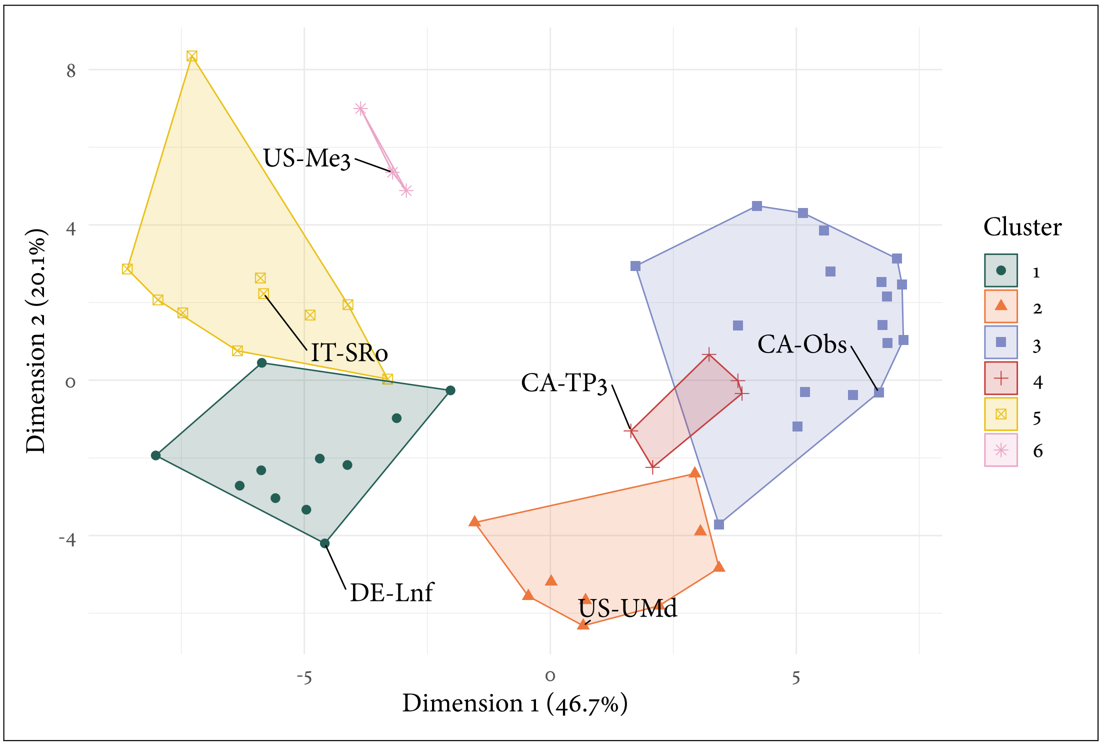
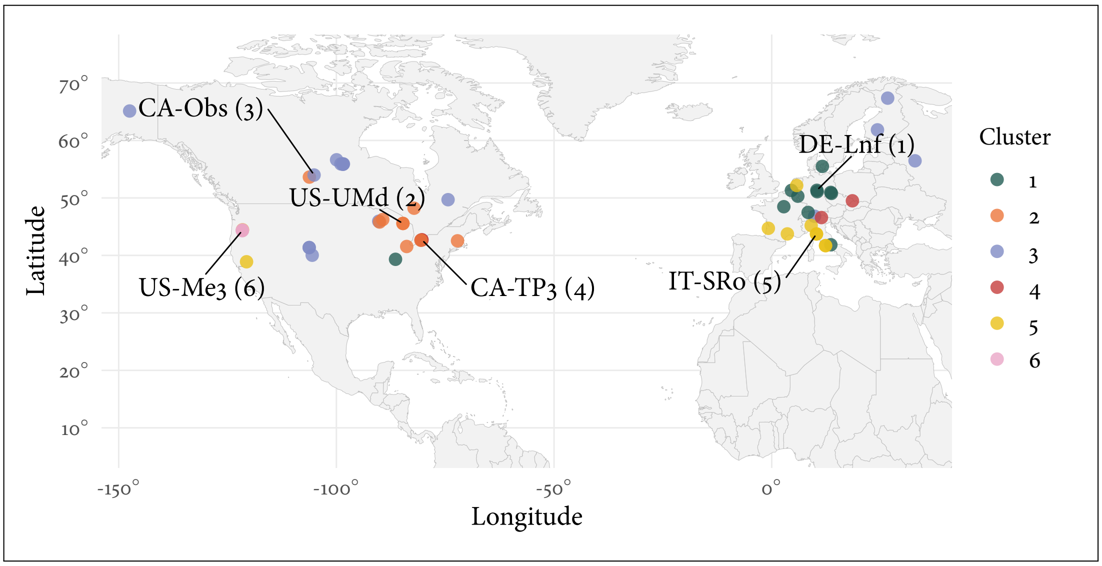
5 FIGURES AND TABLES



(a) Cluster plot indicating the site positions in the space of the first two principal components, with coloured clusters indicating the PAM solution. Axes indicate the % variance explained by each dimension.



(b) Map of all FLUXNET sites and associated clusters in this study, with cluster medoid site IDs highlighted. Visual inspection of cluster geographic distribution aligns with an expected pattern with regards to climatic groupings, suggesting the clustering solution captures climate-relevant patterns.

Figure 5: PAM solution from analysis at $k = 6$ clusters and using Gower's distances shown on a cluster plot in a, and geographic cluster distributions shown in b.

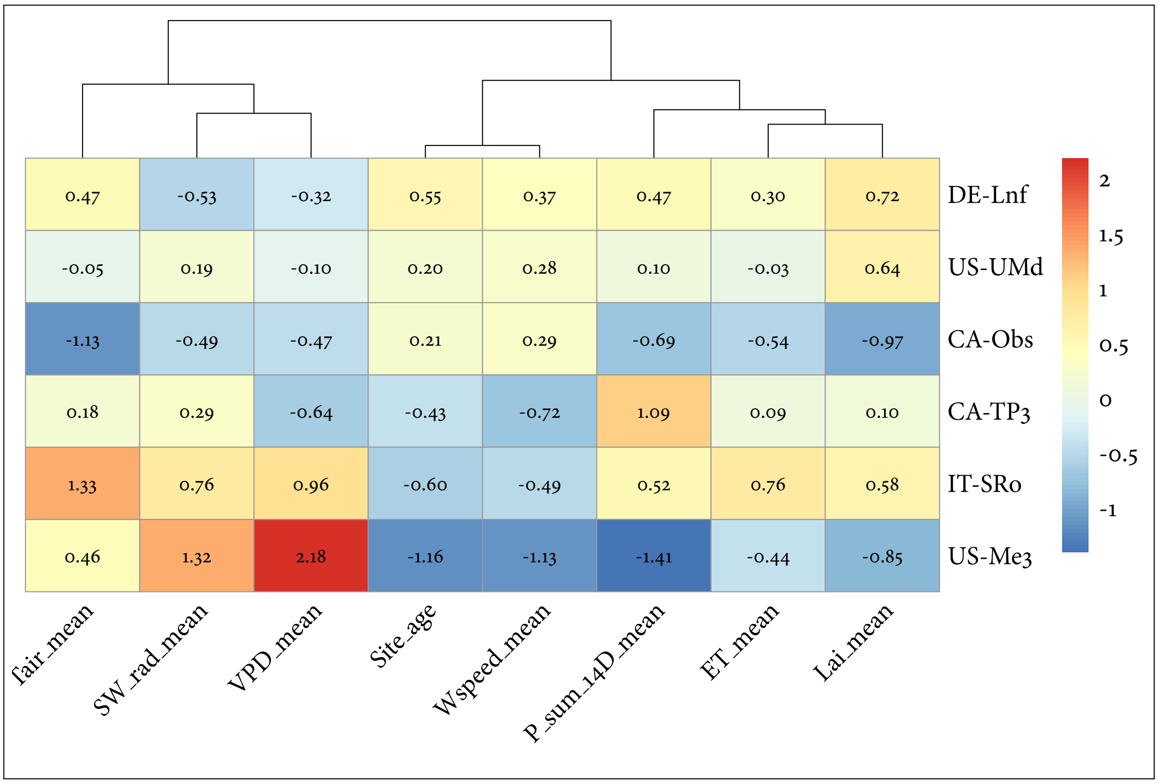


Figure 6: Heatmap of cluster medoid sites and scaled feature values. Scaled features ≥ 1.5 standard deviations from the mean of all sites were classed as 'very' (high or low, depending on direction), while features between 1 and 1.5 standard deviations from the mean were classed as high or low, depending on direction, in order to characterise clusters.

Table 1: Main characteristics of each cluster defined during cluster analysis

Cluster	Medoid Site	Defining Characteristics
1	DE-Lnf	Temperate sites without significant features.
2	US-UMd	Continental sites without significant features.
3	CA-Obs	Continental sites with low mean air temperatures.
4	CA-TP3	Continental sites with high precipitation.
5	IT-SRo	Temperate sites with high air temperatures.
6	US-Me3	Temperate sites with very high VPD, high SW radiation, low stand age, low wind speeds, and low precipitation.

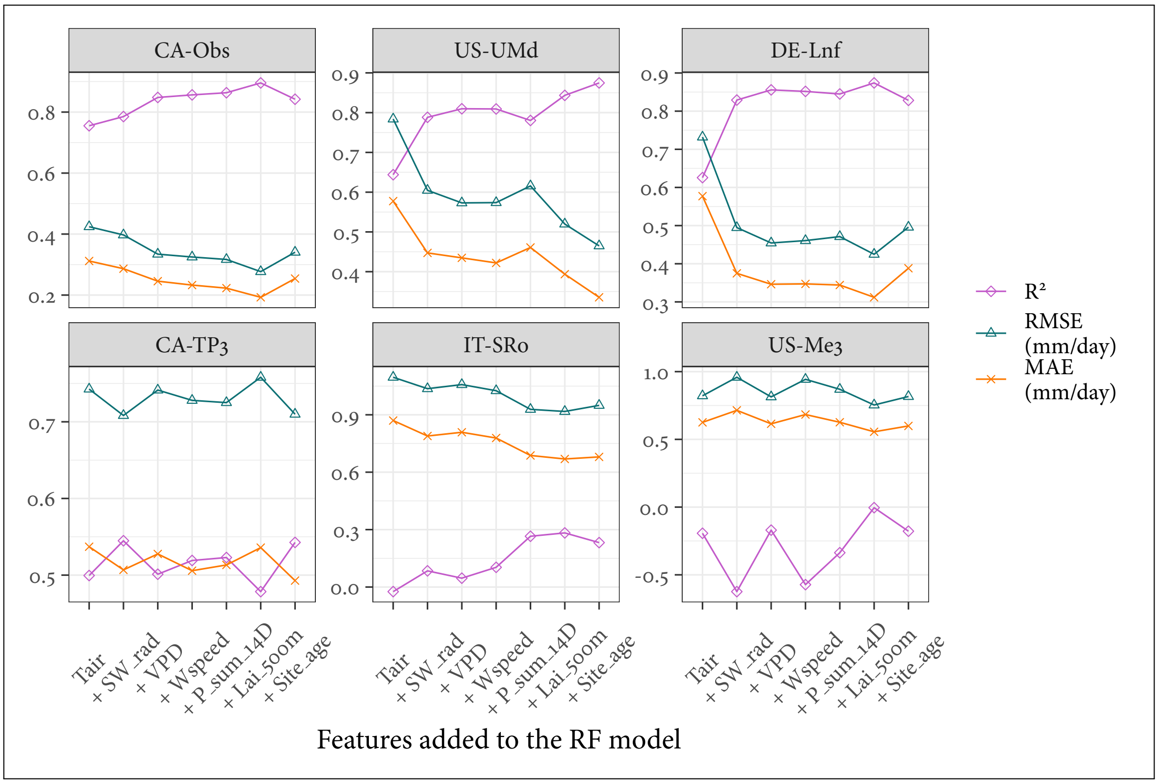


Figure 7: R^2, RMSE, and MAE for medoid cluster site predictions of daily ET during sensitivity testing (ordered left-right and top-bottom by peak R^2). Models were built and tested using a consecutively greater number of input features from FLUXNET and LAI data (for example, the '+VPD model' incorporates Tair, SW_rad, and VPD as training and input features). R^2 performance at only half of the sites was moderate or strong, and suggests that adding features up to LAI improved prediction accuracy at four of the six medoid sites. Adding site age as a final feature led to mixed performance across ET predictions made at medoid sites, with only two of the six benefiting in terms of predictive performance from this input feature.

[space for plot as in Rao et al. [9] (fig. 3, E/F)]

6 SUPPLEMENTARY MATERIALS AND METHODS

All scripts used in this study, alongside a guide of how to re-run the full analysis, can be found at: `https://github.com/archiebenn/forest-age-ET.git`

6.1 Data Acquisition and Setup

Flux tower data were downloaded from the FLUXNET2015 dataset [10] and processed using `R` (version 4.5.3) [11] with the `tidyverse` [12] suite of packages. Flux data were filtered to only keep North American and European sites whose forest ages are dated in Besnard et al. [5], with rows of variables within these sites filtered based on their provided FLUXNET QC flag value. ET rates were calculated row-wise from the provided `LE` (latent heat flux) and `T_air` columns using the `LE.to.ET()` function from `bigleaf` [13] and multiplying this value by 86400 ($\mathrm{s\,d^{-1}}$). Therefore, ET represents actual ET (AET) throughout this study. Four-day LAI data (NASA product: MCD15A3H [14]) were retrieved using site coordinates via `MODISTools` [15], filtered on QC, and linearly interpolated between the four-day periods of LAI to provide a continuous daily value, before joining by date and site to the main FLUXNET dataset.

Site stand ages retrieved from Besnard et al. [5] are static despite multi-year data ranges often contained at these sites in the FLUXNET. As such, site ages were extrapolated across the range of data at each site to act as a continuous metric, with the provided age set to the first row of the FLUXNET data for each site. Köppen climate sites were retrieved and attached based on site latitude and longitude, and these classifications were further simplified to their main groups (Temperate and Continental in this case). A two week cumulative sum of precipitation prior to the day of measurement was also calculated and attached per row, with the first 13 days of data at each site then being discarded.

6.2 Cluster Analysis

Cluster analysis is an exploratory technique which aims to classify objects into as-yet-unknown groups based on their similarities [16]. To group sites in this study using cluster analysis, a Gower's distance matrix was first formed. Gower's distances are a similarity measure which can handle both numerical and categorical data to provide a distance value between two data points (sites) from 0 (completely dissimilar) to 1 (completely similar) [6]. A table of the site features used for these distance calculations can be seen in Table 2, and in this study Gower's distance is treated as a measure of ecological dissimilarity between pairs of sites. The Partitioning Around Medoids (PAM) algorithm and silhouette method was then used to map the Gower's distance matrix into a specified set of clusters [7]. PAM identifies medoids, which are the points within each cluster whose sum of dissimilarities to all other sites in the same cluster is minimal, and so, in this case, can be identified as the most representative flux tower sites of each cluster.

Table 2: Variables and definitions used to calculate Gower's distances during cluster analysis.

Variable	Data Type	Definition
`ET`	Numerical	Mean daily ET rate ($mm\,d^{-1}$)
`T_air`	Numerical	Mean air temperature at site (°C)
`SW_rad`	Numerical	Mean daily incoming short-wave radiation ($W\,m^{-2}$)
`VPD`	Numerical	Mean daily vapour pressure deficit (hPa)
`Site_age`	Numerical	Mean forest stand age at site (years)
`Wspeed`	Numerical	Mean daily wind speed ($m\,s^{-1}$)
`P_sum_14D`	Numerical	Mean two-week cumulative sum of precipitation prior to the day of measurement (mm)
`Lai_500m`	Numerical	Mean daily leaf area index ($m^2\,m^{-2}$)
`Cover_type`	Categorical	IGBP cover type classifiation of site
`Climate_zone`	Categorical	Main Köppen climate group of site

6.3 Random Forest modelling

Random Forest models were trained and tested with `RandomForestRegressor()` from `scikit-learn` [17] using the different methods outlined below. Due to time constraints, a single set of optimised parameter values was calculated once using `GroupKFold()` and `Optuna` [18] to minimise root mean square error (RMSE) on fold predictions. These values were used for all RF models throughout the study (see Table 3). In all cases, regression analysis metrics of R^2 (coefficient of determination), RMSE, and mean absolute error (MAE) were calculated from model predictions of ET using their respective `scikit-learn` functions alongside observed values of ET at each site. R^2 was the favoured metric throughout this study due to the interpretability limitations of RMSE and MAE in regression analysis set out by Chicco, Warrens, and Jurman [19].

Table 3: Parameters selected for use in all RF training setups in this study. Calculated to minimise RMSE on trials using `GroupKFold()` and `Optuna`.

RF parameter	Value
`n_estimators`	350
`max_depth`	17
`min_samples_leaf`	19
`max_features`	0.35

6.3.1 Model Sensitivity Testing

To assess the sensitivity of the RF models to different features, a forward feature selection assessment was carried out. Non-target (ET) numerical features were added sequentially and the RF model was re-trained and tested by forming daily ET predictions at the medoid sites with each new feature added. Features were added in the following order: `T_air`, `SW_rad`, `VPD`, `Wspeed`, `P_sum_14D`, `Lai_500m`, `Site_Age`. Training sites consisted of all sites apart from medoids (see **??**), with predictions formed and testing carried out at medoid sites.

6.3.2 Random Sampling of Training Datasets

To investigate how predictive performance is impacted by including site age as a feature, two separate RF architectures were formed: 'Mod' (Equation 1) and 'Mod + Age' (Equation 2). To further assess how the overall ecological similarity of training dataset composition to the test site may impact predictive accuracy of daily ET, a method to subset sites from a larger site pool before RF training was also developed (script `21_analysis_1.2.py`), with details on how this analysis runs below.

Sites were first separated as follows: 6 medoid sites for testing and the remaining 44 sites as the full training pool. Following this, 20 sites were chosen at random (`NumPy.random` module) from the training pool, and then two models - 'Mod' and 'Mod + Age' - are trained on this subset of sites. After training, both models then predict on the six held out medoid sites, and the mean Gower's distance from the full training subset to each test site is calculated and stored alongside prediction performance metrics. This paired process is then repeated a further 99 times, each with a new `NumPy` seed to avoid using the same subset, to produce 100 runs using this method.

$$ET_i = RF_{Mod}(T\_air_i, SW\_rad_i, VPD_i, W\_speed_i, \sum_{i-13}^{i} P_i, LAI_i) \qquad (1)$$

$$ET_i = RF_{Mod+Age}(T\_air_i, SW\_rad_i, VPD_i, W\_speed_i, \sum_{i-13}^{i} P_i, LAI_i, Site\_Age_i) \qquad (2)$$

where the target ET_i is the estimated ET rate on the ith day (mm/day); $RF(\cdot)$ is the Random Forest model; the features T_air, SW_rad, VPD, W_speed, $\sum_{i-13}^{i} P_i$, LAI, $Site\_Age$ are all listed with descriptions in Table 1.

6.3.3 Time Series Decomposition

6.4 A Two Site Case Study

6.5 Site Metadata

Table 4: Metadata summary of FLUXNET sites used in this study (*** denotes a cluster medoid)

FLUXNET ID	Days of Data	Country	Climate Zone	Cover Type	Site Age (years)	Mean ET 90D Peak ($mm\,d^{-1}$)
BE-Bra	3590	Belgium	Temperate	MF	92	2.02
BE-Vie	4466	Belgium	Temperate	MF	107	2.15
CA-Gro	3639	Canada	Continental	MF	84	2.45
CA-Man	2290	Canada	Continental	ENF	173	1.90
CA-NS1	1141	Canada	Continental	ENF	157	1.59
CA-NS2	960	Canada	Continental	ENF	76	1.75
CA-NS3	1141	Canada	Continental	ENF	42	1.70
CA-NS5	1137	Canada	Continental	ENF	26	1.88
CA-NS6	1141	Canada	Continental	OSH	17	1.35
CA-NS7	1141	Canada	Continental	OSH	8	1.73
CA-Oas	3029	Canada	Continental	DBF	91	2.82
CA-Obs***	3029	Canada	Continental	ENF	122	1.95
CA-Qfo	2687	Canada	Continental	ENF	106	1.75
CA-TP3***	4550	Canada	Continental	ENF	44	2.45
CA-TPD	1082	Canada	Continental	DBF	100	2.31
CH-Lae	3782	Switzerland	Temperate	MF	190	4.31
CZ-BK1	3920	Czechia	Continental	ENF	37	1.73
DE-Hai	2720	Germany	Temperate	DBF	260	2.68
DE-Lnf***	2670	Germany	Temperate	DBF	122	2.60
DE-Obe	2463	Germany	Temperate	ENF	79	2.21
DE-Tha	4502	Germany	Temperate	ENF	131	2.14
DK-Sor	4482	Denmark	Temperate	DBF	98	2.87
FI-Hyy	4466	Finland	Continental	ENF	59	2.15
FI-Sod	4454	Finland	Continental	ENF	91	1.87
FR-Fon	3586	France	Temperate	DBF	166	3.64
FR-LBr	2346	France	Temperate	ENF	44	2.76
FR-Pue	4550	France	Temperate	EBF	73	1.86
IT-Col	2908	Italy	Temperate	DBF	195	2.58
IT-Cp2	1082	Italy	Temperate	EBF	65	2.38
IT-Cpz	2359	Italy	Temperate	EBF	65	2.02
IT-PT1	897	Italy	Temperate	DBF	15	3.51
IT-Ren	3253	Italy	Continental	ENF	199	3.53
IT-SR2	716	Italy	Temperate	ENF	65	2.21
IT-SRo***	3819	Italy	Temperate	ENF	63	2.51
NL-Loo	4498	Netherlands	Temperate	ENF	119	2.64
RU-Fyo	4458	Russia	Continental	ENF	247	2.51
US-Blo	1901	U.S.A	Temperate	ENF	21	4.04
US-GBT	898	U.S.A	Continental	ENF	181	2.66
US-GLE	3638	U.S.A	Continental	ENF	190	2.36
US-Ha1	3262	U.S.A	Continental	DBF	112	2.40
US-MMS	4538	U.S.A	Temperate	DBF	105	3.23
US-Me3***	2012	U.S.A	Temperate	ENF	23	1.91
US-NR1	4550	U.S.A	Continental	ENF	121	2.71
US-Oho	3615	U.S.A	Continental	DBF	55	4.02
US-PFa	4522	U.S.A	Continental	MF	164	2.35
US-Prr	1355	U.S.A	Continental	ENF	101	1.56
US-Syv	2723	U.S.A	Continental	MF	307	2.52
US-UMB	4538	U.S.A	Continental	DBF	102	3.16
US-UMd***	2715	U.S.A	Continental	DBF	94	2.91
US-WCr	2931	U.S.A	Continental	DBF	106	2.56